# MUON COOLING IN THE RFOFO RING COOLER

R.C. Fernow, J.S. Berg, J.C. Gallardo*, R.B. Palmer, BNL, Upton, NY 11973, USA

*Abstract*

The performance of the ring described here compares favorably with the linear cooling channel used in the second U.S. Neutrino Factory Study[1]. The 6D phase space density of an idealized ring is increased by a factor of 238, compared with the linear channel's factor of only 15. The simulations make use of fully realistic magnetic fields, and include absorber and rf cavity windows, and empty lattice cells for injection/extraction.

## INTRODUCTION

In the present U.S. Neutrino Factory design [2] the muon beam 6D phase space volume must be reduced in order to be able to accelerate it and inject into the storage ring pointing to a long distance neutrino detector. Ionization cooling is currently the only feasible option for cooling the beam within the muon lifetime ($\tau_0 = 2.19\,\mu$s). If muons alternately pass through a material absorber, and are then reaccelerated, and if there is sufficient focusing at the absorber, then the muon's transverse phase space is reduced, *i.e.* the muons are cooled in the transverse dimension. A consequence of the transverse cooling is an increase of the longitudinal phase space caused by energy straggling in the material. The consequent momentum spread can be reduced if dispersion is introduced and a wedge absorber placed such that high momentum particles pass through more material than low momentum particles. However, in this process the beam width is increased. The process is thus primarily an exchange of emittance between the longitudinal and transverse dimensions, but when combined with transverse cooling in the material, all three dimensions can be cooled.

The ring design parameters are given in Tb. 1. All cells are strictly identical, and the presence of an integer betatron resonance within the momentum acceptance is eliminated.

Figure 1 shows a detailed view of three cells of the lattice, in plan (a) and side (b) views. The solenoids are not evenly spaced; those on either side of the absorbers are closer in order to increase the focusing at the absorber. The longitudinal field on-axis has an approximately sinusoidal dependence on position. The beam axis is displaced laterally with respect to the coil centers (as shown in Fig. 1a) to minimize horizontal fields that cause vertical beam deviations.

The bending field for the ring, and the required dispersion, are provided by alternately tilting the solenoids by $3.0°$.

Table 1: RFOFO Basic Ring Parameters

| | |
|---|---|
| Circumference (m) | 33 |
| Cells | 12 |
| Max $B_z$ (T) | 2.77 |
| Coil Tilts (deg.) | 3.0 |
| Ave Momentum (MeV/c) | 220 |
| Min Trans. Beta (cm) | 38 |
| Max. Dispersion (cm) | 8 |
| Momentum Compaction | 0.0037 |
| Wedge Absorber Material | $H_2$ |
| Wedge Thickness on axis (cm) | 25.4 |
| Wedge Angle (°) | 90 |
| Wedge Vertex position (cm) | 12.7 |
| Wedge Azimuthal angle (°) | 230 |
| Frequency (MHz) | 201.25 |
| Gradient (MV/m) | 12 |
| Phase (°) from 0-crossing | 25 |

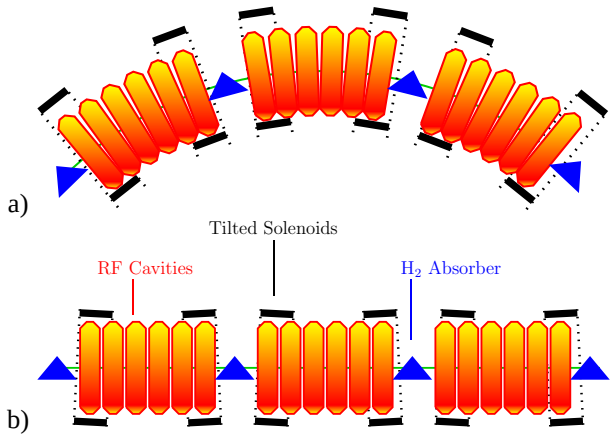



Figure 1: Three cells of the RFOFO lattice; a) plan view; b) side view. Notice that the coils have been displaced radially by 10 cm.

## MODELING THE RING

The RFOFO ring was modeled using the ICOOL code[3]. The magnetic field from the tipped solenoids was calculated in an independent code that found the resultant field by summing the fields from a system of current sheets[4, 5]. The field of a solenoidal current sheet can be written analytically in terms of elliptic integrals. The resultant field components were shown to satisfy the 3D Maxwell equations to a high level of accuracy and agreed with independent calculations[6, 7]. The solenoidal field on-axis was approximately sinusoidal with a peak magni-

* Correspondent: gallardo@bnl.gov

tude of 2.7 T. The solenoids were tipped to produce an average vertical dipole field of 0.125 T on-axis. The circle containing the centers of the solenoids was displaced 10 cm outwards radially from the nominal beam axis in order to minimize the horizontal dipole field on the axis.

The rf cavities were modeled using cylindrical pillboxes running in the TM010 mode. The cavities are located in a dipole field in this design. Since we are accelerating muons, the cavities can be enclosed with metallic end windows in order to produce the maximum electric field on-axis for a given amount of rf power. The rf windows were stepped in thickness radially, in order to provide minimum thickness near the beam axis and still control the temperature increase due to rf heating.

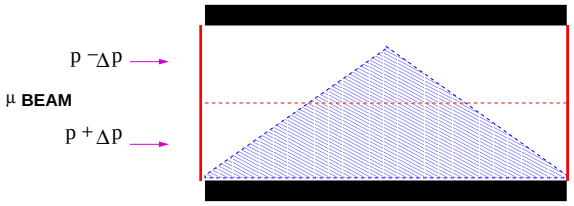



Figure 2: Wedge absorber used in the simulations.

The *wedge* absorbers are house-shaped, as shown in Fig. 2. The wedge only extends part way across the gap, such that a particle on the reference orbit always loses the same amount of energy crossing the wedge. The absorber windows were planar and located axially just in front of and behind the wedge itself. This is likely a worst case because the whole beam is forced to cross the window regardless of its transverse position. In reality the window shape will conform to the absorber and the effect on the beam of scattering in the window should be lessened.

We use a Gaussian input beam[6] in the simulations with a normalized transverse emittance of 12 mm and a normalized longitudinal emittance of 18.4 mm. The reference momentum is 203 MeV/c. This value was chosen to make the circulation time around the ring on a closed orbit equal to the 25th harmonic of 201.25 MHz. The initial beam has a correlation between the axial momentum and the transverse amplitude, to minimize the tendency for the particles in the bunch to spread out longitudinally in the solenoidal field. The initial *rms* bunch length is 8 cm, so this ring is not compatible with the bunch train assumed in Study-2. The use of compact cooling rings will likely also require the use of a preceding bunch compression ring.

The mean transverse position of the starting bunch was displaced by -10 mm in $x$ and 18 mm in $y$ to put the reference momentum particle on a closed orbit.

## SIMULATION RESULTS

Three merit factors have been widely used in the discussion of cooling ring performance. The quantity **M-factor**, defined by $M = \frac{\epsilon_6(\text{initial})}{\epsilon_6(\text{final})} \times$ Transmission, is most useful for collider applications. Another figure of merit, which is local in character, is the **Q-factor** defined by $Q(s) = \frac{\frac{d\,\epsilon_6(s)}{ds}}{\frac{d\,N(s)}{ds}} \frac{N(s)}{\epsilon_6(s)}$; if $Q(s)$ is constant, then $< Q >= Ln(\frac{\epsilon_6(s)}{\epsilon_6(0)})/Ln(\frac{N(s)}{N(0)})$. The third merit factor, and the one most directly useful for neutrino factories, is the muon density into a fixed accelerator acceptance volume. If $N$ is the number of muons and V is a fixed acceptance volume in phase space, then we define the **D-factor** at any location $s$ along the ring to be $D(s) = \frac{N(s)/V}{N(0)/V}$.

We first consider the performance of an idealized ring before the introduction of windows around the absorbers and rf cavities and before the introduction of empty cells for injection/extraction. Figure 3 shows the three emittances, transverse, longitudinal and 6D, merit factor M and transmission plotted on a logarithmic scale versus distance traveled. Table 2 summarizes the performance. The **Q-factor** is 8.7. After a distance of 400 m ($\approx$ 12 turns), the

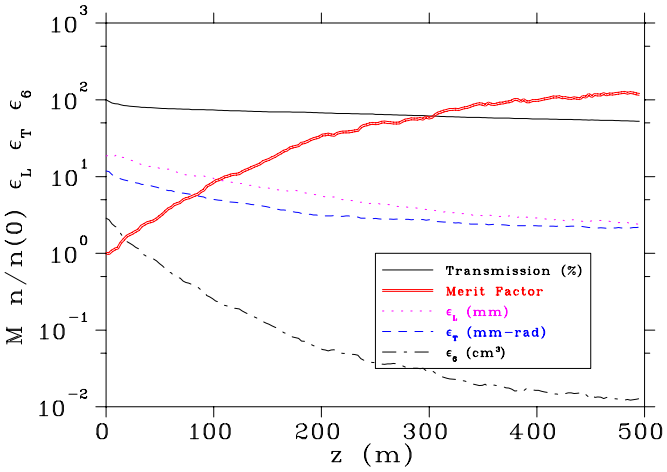



Figure 3: Performance of RFOFO Ring. Transmission, normalized transverse emittance, normalized longitudinal emittance, normalized 6-dimensional emittance, and the merit factor M, as a function of distance.

6D emittance has fallen by a factor of 238, with a transmission of 53% (66% without decay). The **M-factor** is 125. The same factor for the Study-2 cooling lattice, also without windows, is 15.

Table 2: RFOFO Ring Performance

| | Before | After | ratio |
|---|---|---|---|
| $\epsilon_\perp$ ($\pi$mm$^2$) | 11.87 | 2.14 | 5.5 |
| $\epsilon_\parallel$ ($\pi$mm) | 18.64 | 2.53 | 7.4 |
| $\epsilon_6$ ($\pi^3$cm$^3$) | 2.86 | 0.012 | 238 |
| $N/N_0$, inc. decay | 1 | 0.53 | 0.53 |
| **M-factor** | 1 | 125 | **125** |
| **D-factor** | 1 | 8.57 | **8.57** |

We next show the idealized ring behavior in terms of the

muon density. Figure 4 shows the total muon transmission together with the muon density into two fixed acceptance volumes. These volumes could correspond to the acceptance of a linear accelerator that follows the cooling ring at a neutrino factory.

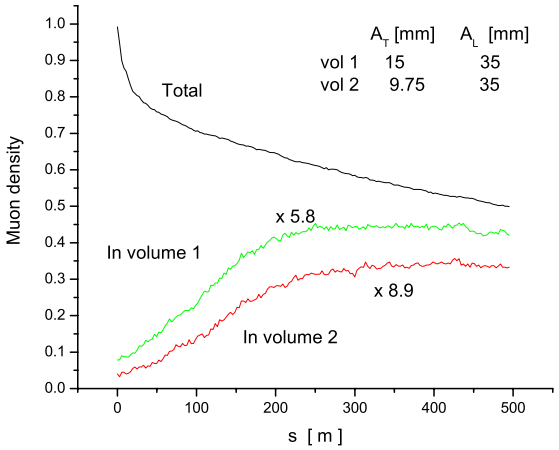



Figure 4: Performance of RFOFO Ring. Transmission and muon density into two fixed acceptance volumes. These results were obtained using a $100^\circ$ wedge.

The idealized ring increases the muon density into the smaller acceptance volume by a factor of almost 9 in 250 m, which corresponds to about 8 turns. The density in the larger acceptance volume increases by about a factor of 6.

The introduction of windows around the absorbers or rf cavities introduces new sources of scattering and degrades the cooling performance. In addition a real ring needs space for injection/extraction. This is treated here by leaving the rf and absorbers out of two of the 12 lattice cells. The empty cells have the same pairs of solenoids however, so the magnetic field periodicity is preserved. Table 3 shows the result of a) adding absorber windows, b) adding rf cavity windows, and c) leaving empty cells in the lattice.

The absorber window used in Study-2 was 360 $\mu$m of aluminum. We see that windows of this thickness degrade the performance by about $30\%$. For safety reasons it may be necessary to use an additional window that increases the total amount of aluminum. On the other hand the use of other materials or optimized window shapes could reduce the amount of material. Another possibility would be to replace the liquid $H_2$ absorber with a solid material. LiH is one possible candidate, although Tb. 3 shows there is a $45\%$ loss of performance with this option.

The rf Be windows degrade the performance by about $35\%$. One possibility to get around this problem is to operate the cavities at liquid nitrogen temperature. The lower operating temperature and the reduced rf gradient of 12 MV/m versus 16 MV/m in Study-2 allow the thickness of the windows to be decreased by a factor of 20 and the performance loss is only $13\%$. Alternatively, one could eliminate the rf cavity windows altogether and use an open cavity. This has the disadvantage that four times more power is required to produce the same $E_z$ on axis.

Table 3: Perturbations on the idealized ring performance. FS2 stands for Study-2 used windows; FS2/20 stands for windows used in Study-2 with thicknesses reduced by a factor of 20.

| absorber | abs win | rf win | empty cells | $D$ |
|---|---|---|---|---|
| LH | none | none | 0 | 8.93 |
| LH | 250$\mu$m AL | none | 0 | 7.50 |
| LH | 360$\mu$m AL | none | 0 | 6.60 |
| LH | 500$\mu$m AL | none | 0 | 6.08 |
| LiH | none | none | 0 | 4.88 |
| LH | none | FS2 | 0 | 5.88 |
| LH | none | FS2/20 | 0 | 7.80 |
| LH | none | none | 2 | 6.73 |
| LH | 360$\mu$m AL | FS2/20 | 2 | 4.25 |

Introducing empty cells for injection/extraction reduces the performance by $25\%$.

Finally, we consider an example that combines all of these effects. We choose liquid $H_2$ as the absorber with Study-2 like windows, assuming we can use this effective amount of aluminum because of improved alloys or window shape. We assume operation at liquid nitrogen temperature and use the thinner Be rf windows and leave empty cells in the lattice for injection/extraction. This realistic ring model still gives an impressive increase in the muon density of a factor of 4.25

## REFERENCES

[1] S. Ozaki, R. Palmer, M. Zisman, J. Gallardo, Editors, *Feasibility Study II of a Muon Based Neutrino Source*, BNL-52623, June, 2001.

[2] The MC Collaboration, *Recent Progress in Neutrino Factory and Muon Collider Research within the Muon Collaboration*, to be published. (http://www.cap.bnl.gov/mumu/pubs/prstab-030422/prstab.pdf)

[3] R. Fernow, *ICOOL: a simulation code for ionization cooling of muon beams*, Proc. 1999 Particle Accelerator Conference, p. 3020-3022.

[4] R. Fernow, J. Gallardo, *Realistic on-axis fields for the RFOFO cooling ring*, MUC-NOTE-COOL-THEORY-265, Nov. 2002. This series of technical notes can be found at (http://www-mucool.fnal.gov/mcnotes/)

[5] R. Fernow, J. Gallardo, *Calculation of RFOFO fields using the off-axis expansion in ICOOL*, MUC-NOTE-COOL-THEORY-268, Jan. 2003.

[6] V. Balbekov, *Simulation of RFOFO ring cooler with tilted solenoids*, MUC-NOTE-COOL-THEORY-264, Nov. 2002.

[7] S. Bracker, *Magnetic field maps for the RFOFO muon cooling ring*, MUC-NOTE-COOL-THEORY-271, Mar. 2003.